

Periodontology 1

Periodontal Assessment

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Learning Objectives

1. Understand and describe the components of a Periodontal Assessment and Examination
2. Understand how anatomy, biofilm, pathogenesis and host response influence the signs we see clinically
3. Understand probing depth vs. clinical attachment level/loss and the difference
4. Understand primary and secondary etiology and give examples of each
5. Understand mobility, furcation, and plaque indices used

Clinical Assessment of Periodontal Conditions

- How do we know if our patient has experienced tissue destruction/loss as the result of pathogenesis?
- Evaluate signs and symptoms
- Gather information, classify and diagnose periodontal condition, prognosis, treatment plan

Classification

Periodontal Health, Gingival Diseases and Conditions

- Periodontal Health and Gingival Health
- Gingivitis: Dental Biofilm-Induced
- Gingival Diseases: Non-Dental Biofilm-Induced

Periodontitis

- Necrotizing Periodontal Diseases
- Periodontitis
- Periodontitis as a Manifestation of Systemic Disease

Periodontal Manifestations of Systemic Diseases and Developmental and Acquired Conditions

- Systemic Diseases or Conditions Affecting Periodontal Supporting Tissues
- Mucogingival Deformities and Conditions
- Traumatic Occlusal Forces
- Tooth- and Prosthesis-Related Factors
- Periodontal Abscesses and Endodontic-Periodontal Lesions

Peri-Implant Diseases and Conditions

- Peri-Implant Health
- Peri-Implant Mucositis
- Peri-Implantitis
- Peri-Implant Soft and Hard Tissue Deficiencies

Signs vs. Symptoms



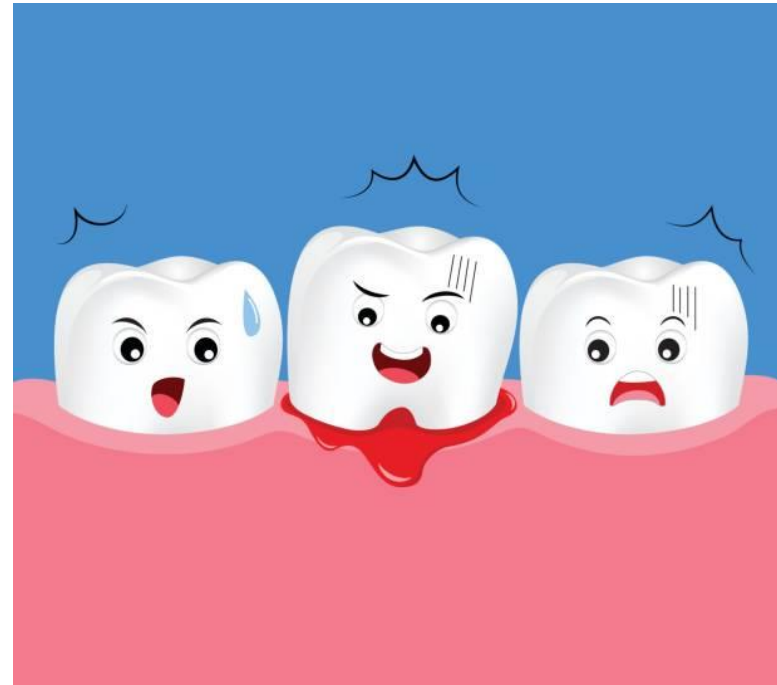
Signs – Definitive, objective evidence from a physician or dentist that shows disease.



Symptoms – Subjective descriptions from a patient on how he or she is feeling or what he or she is experiencing.

Assessment of periodontal condition:

1. **Symptoms of patient**
2. **Medical and Dental History**
3. **Visual Assessment**
4. **Periodontal Examination**
5. **Radiographic Assessment**
6. **Identification of etiology**



1. Patient Symptoms for Periodontal Conditions

- Loose teeth/spaces
- Bleeding gums
- Swollen/receded gums
- Halitosis or “bad breath”
- Pain or “itchy” gums
- **No symptoms-Periodontitis is often painless until late stages**



2. Medical and Dental History

- Systemic conditions
- Medications
- Previous dental treatment (periodontal, orthodontic, prosthetic)
- Personal /Social factors/Goals
- Current/future dental treatment plans



Dental History

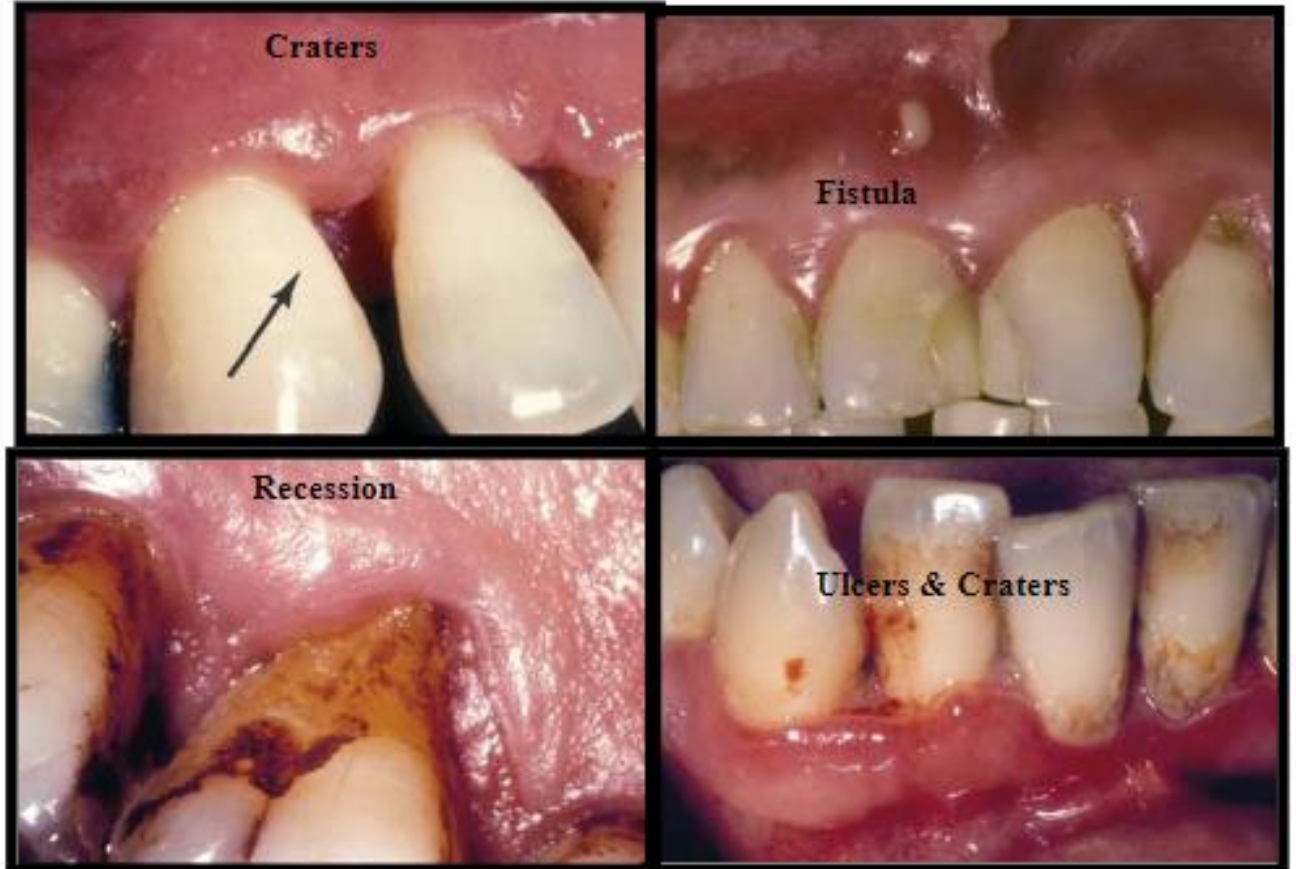
- Presence of other dental issues:
 - Caries, edentulism, bruxism, crowding, poor esthetics, neglect
 - Risk, Dental "IQ", Motivation
- Previous treatment:
 - Periodontal? 4 W's (Who, where..)
 - Prosthetic/Restorative
 - Orthodontic?



3. Visual Assessment

- Gingival description
- Abnormalities
- General dental condition
 - Decay/Fractured teeth
 - Missing teeth
 - Occlusion
 - Existing prostheses

Examples of visual detection of clinical signs of periodontal damage.



4.Periodontal Examination

- 1.Probing depths
- 2.Plaque index
- 3.Bleeding on probing/exudate
- 4.Recession
- 5.Clinical attachment level
- 6.Tooth mobility/ fremitus
- 7.Furcation involvement
8. Occlusion



Periodontal Examination: Axium

1. **P Depth**: Probing depths
2. **GM CEJ**: Gingival margin to CEJ
(+,0, or -) *recession +, over CEJ -*
3. **Attach**=Clinical attachment level
or loss (calculation-indirect)
4. **Calculus** (C or N)
5. **Plaque** (P or N): O'Leary PI
6. **Suppuration**
7. **Bleeding on probing** (B or N)
8. **Furcation involvement** (1,2,3)
9. **Mobility** (0,1,2,3)

3 1 2	3 2 2	3 1 2	2 1 2	2 1 2	2 1 2	2 1 2		P Depth
-3 -1 -2	-3 1 -2	-3 -1 -2	-2 -1 -2	-2 -1 -2	-2 -1 -2	-2 -1 -2		GM CEJ
0 0 0	0 3 0	0 0 0	0 0 0	0 0 0	0 0 0	0 0 0		Attach
N N N	N N N	N N N	N N N	N N N	N N N	N N N	N N N	Calculus
P P P	P P P	P P P	P P P	P P P	P P P	P P P	N N N	Plaque
0 0 0	0 0 0	0 0 0	0 0 0	0 0 0	0 0 0	0 0 0	0 0 0	Suppuratn
N N N	N N N	N N N	N N N	N N N	N N N	N N N	N N N	Bleed
					0	0	0	Furcation
9	10	11	12	13	14	15	16	
3 3 3	3 1 3	3 1 2	2 1 3	2 1 3	3 2 3	2 1 2		P Depth
-3 -3 -3	-3 -1 -3	-3 -1 -2	-2 -1 -3	-2 -1 -3	-3 -2 -3	-2 -1 -2		GM CEJ
0 0 0	0 0 0	0 0 0	0 0 0	0 0 0	0 0 0	0 0 0		Attach
N N N	N N N	N N N	N N N	N N N	N N N	N N N	N N N	Calculus
P P P	P P P	P P P	P P P	P P P	P P P	P P P	N N N	Plaque
B N N	N N N	N N B	N N N	N N N	N N N	N N N	N N N	Bleed
					0 0	0 0	0 0	Furcation
0	0	0	0	0	0	0	0	Mobility

P Depth: Probing Depth

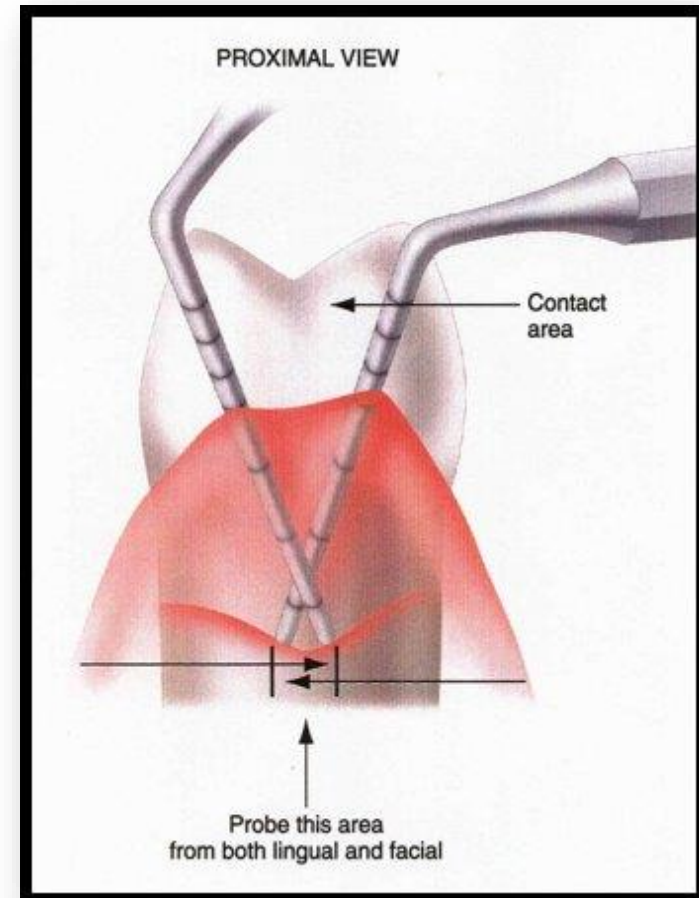
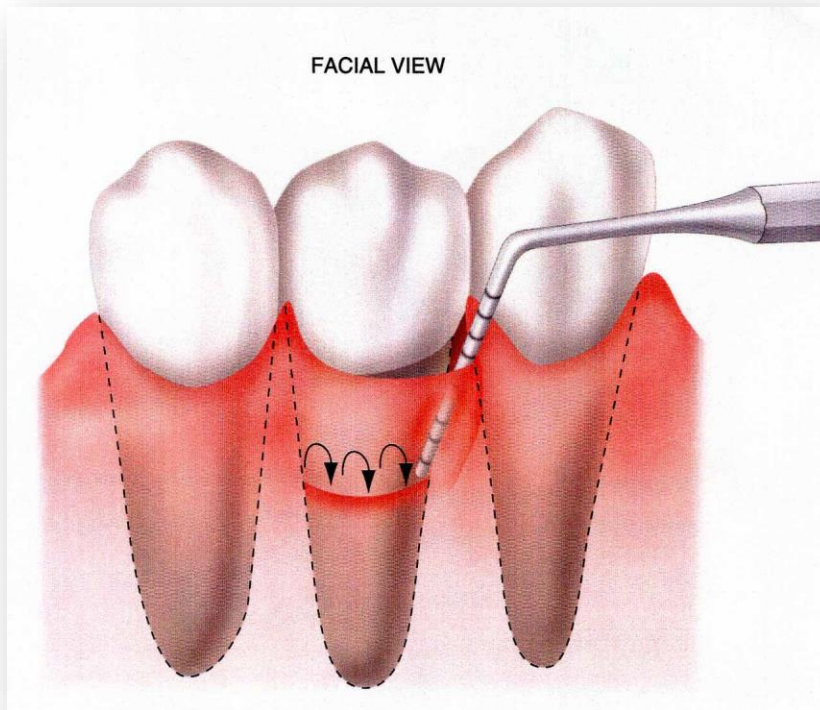


UNC Probe



Marquis
Probe

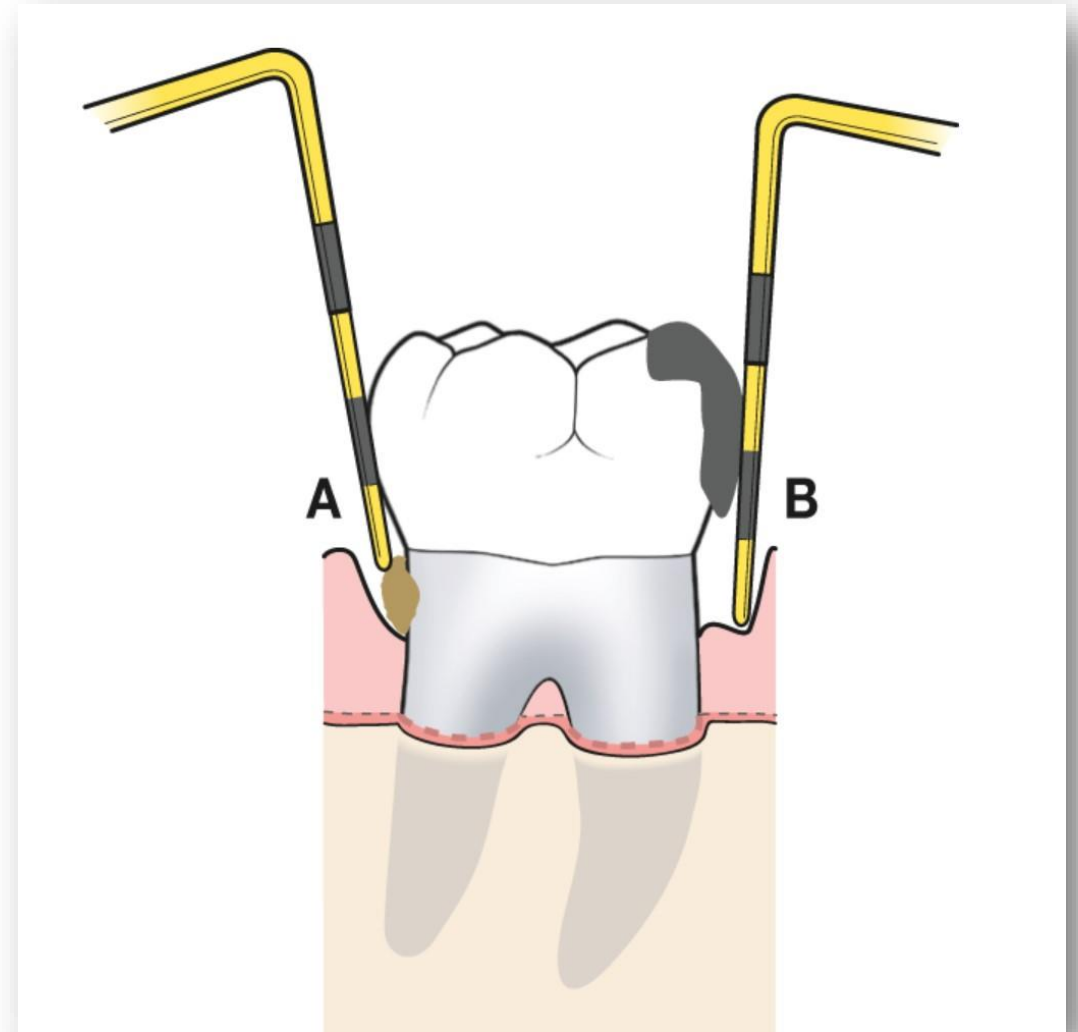
Probing technique



Periodontal probing involves “stepping” a calibrated probe around the entire tooth as close to its long axis as possible

Errors in PD

- Angulation and adaptation
- Pressure (too much , too little)
- Reading the probe
- Patient discomfort
- Interferences: calculus, restorations
- Variance between operators =/- 1mm
- Deeper PD= less accurate
- Diameter/markings of probe



5mm Standard

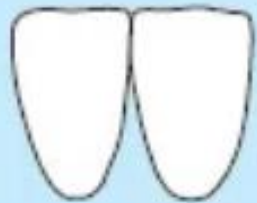
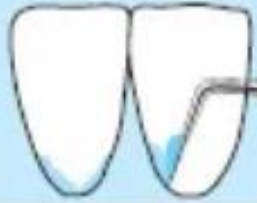
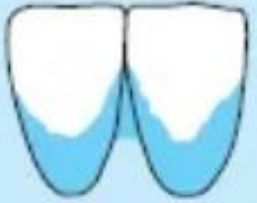
- Why is 5mm "bad"??

5mm Probing
Depth

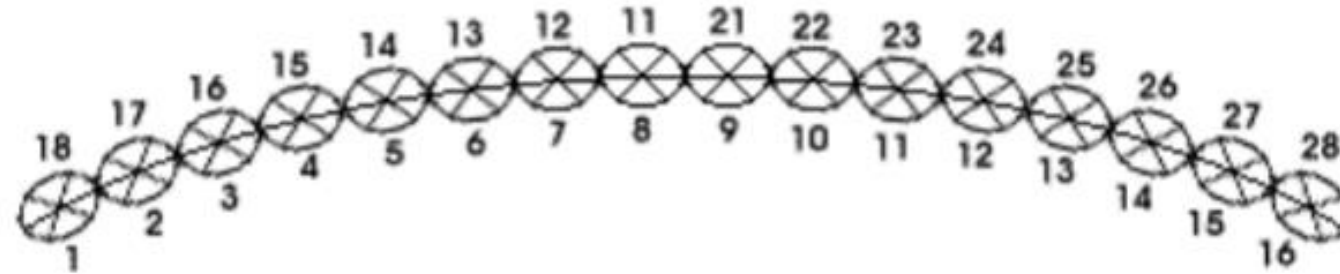
5mm Clinical
Attachment
Loss



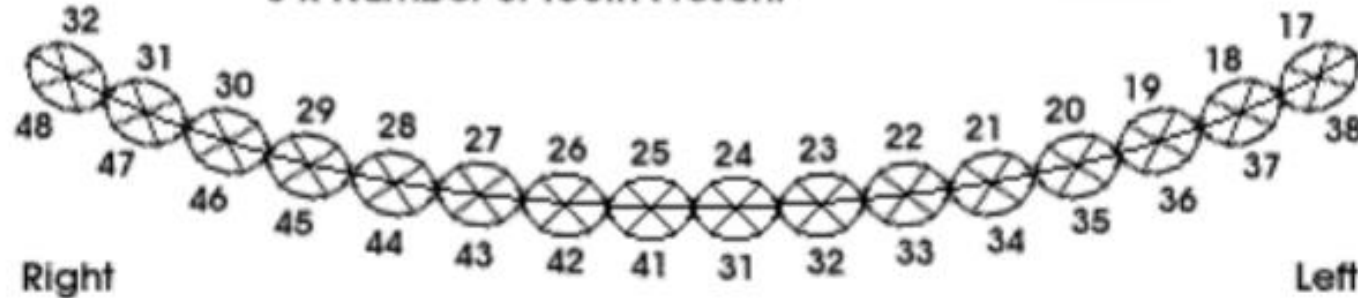
Plaque Index (Silness & Loe 1964)

Grade 0	No Plaque		
1	Thin plaque layer at the gingival margin, only detectable by scraping with a probe		
2	Moderate layer of plaque along the gingival margin; interdental spaces free, but plaque is visible to the naked eye		
3	Abundant plaque along the gingival margin; interdental spaces filled with plaque		
			Abbreviation Grade Pl 0-3

O'Leary Plaque Index



$$\frac{\text{Total Number Tooth Parts With Plaque}}{6 \times \text{Number of Teeth Present}} \times 100 = \text{ } \%$$



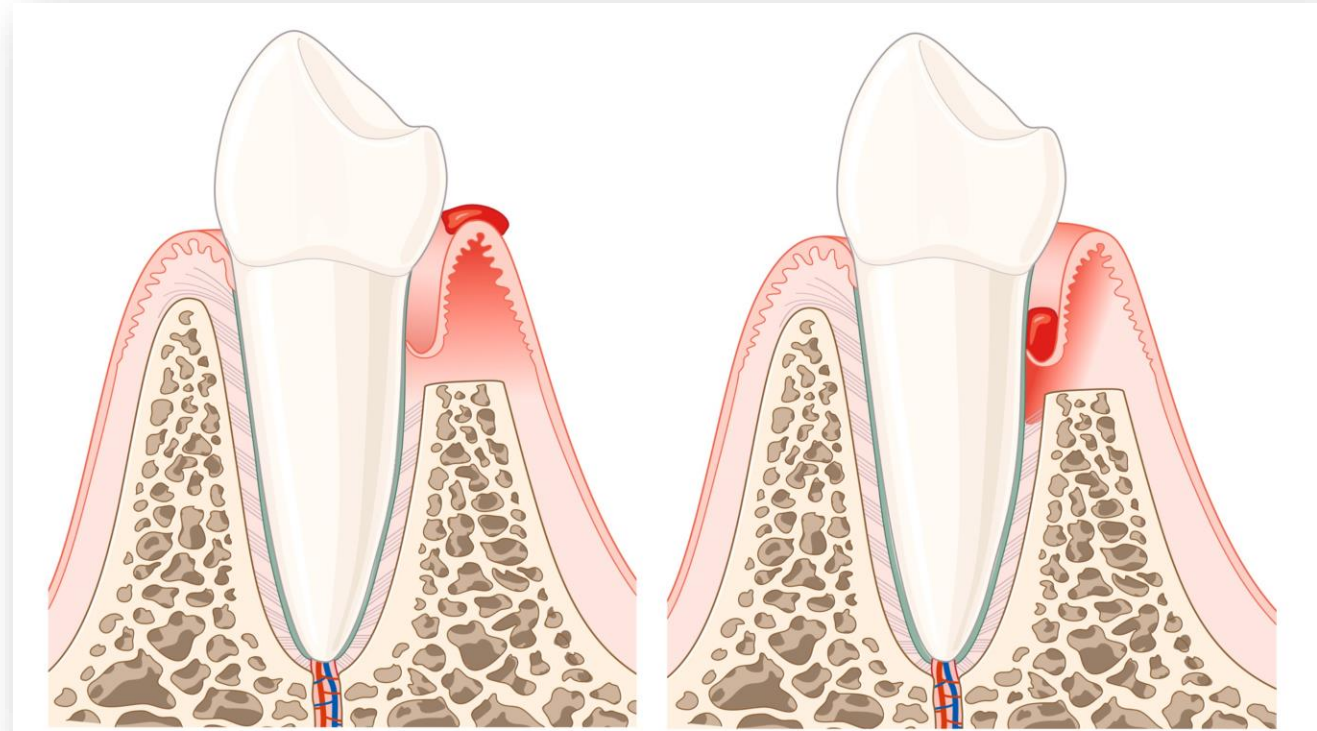
Date: _____

Total Plaque Surfaces _____ / Total Surfaces _____

= O'Leary: _____ % BI: _____ %

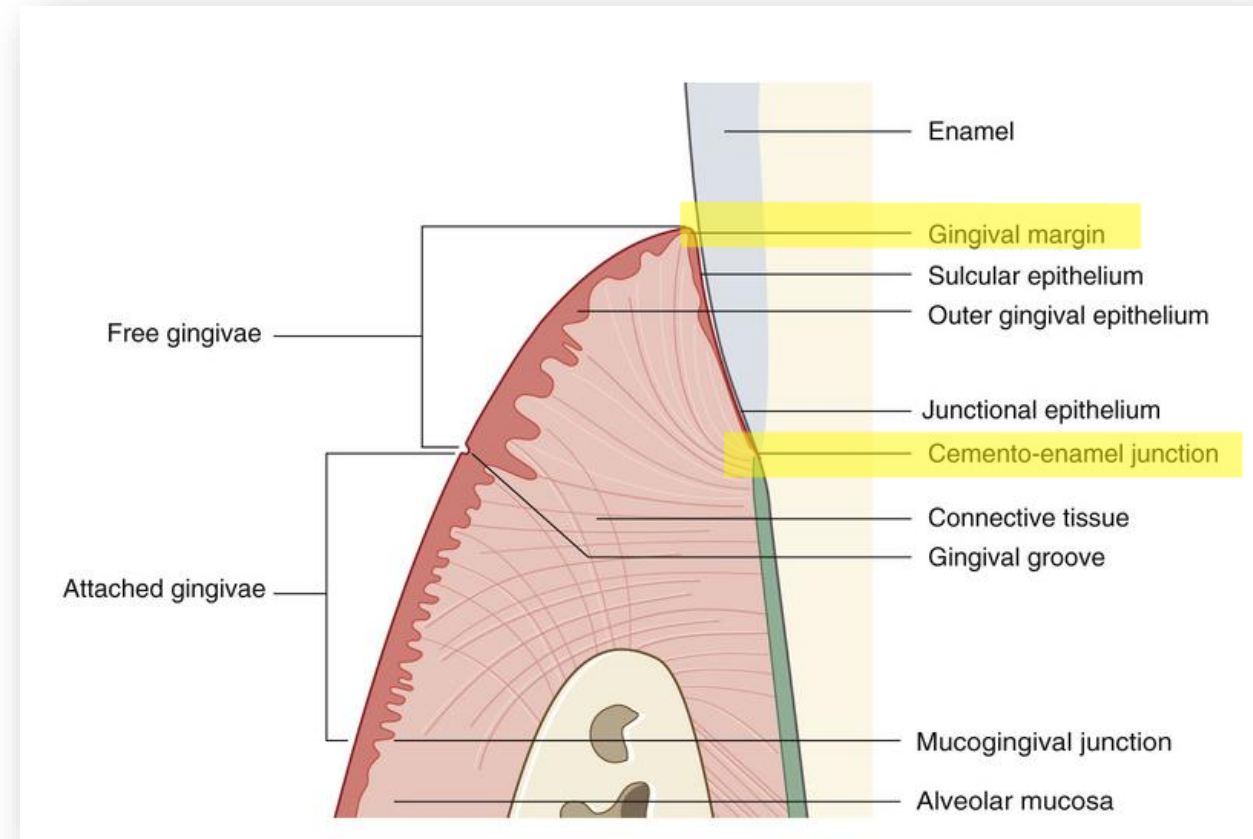
Bleeding on Probing “Bleed”

- Sign of active inflammation
- Atrophic ulcerated pocket epithelium
- Can occur from GM or base of PD
- Indication of progression of disease
- Its absence implies stability



GM to CEJ

- In health: GM is 1-3mm over the CEJ
- To determine GM to CEJ in a patient with Health, Gingivitis or Gingival Enlargement need to measure distance between GM and CEJ
- CEJ is “buried”... To detect it requires an explorer because you can’t see it (“tactile”)



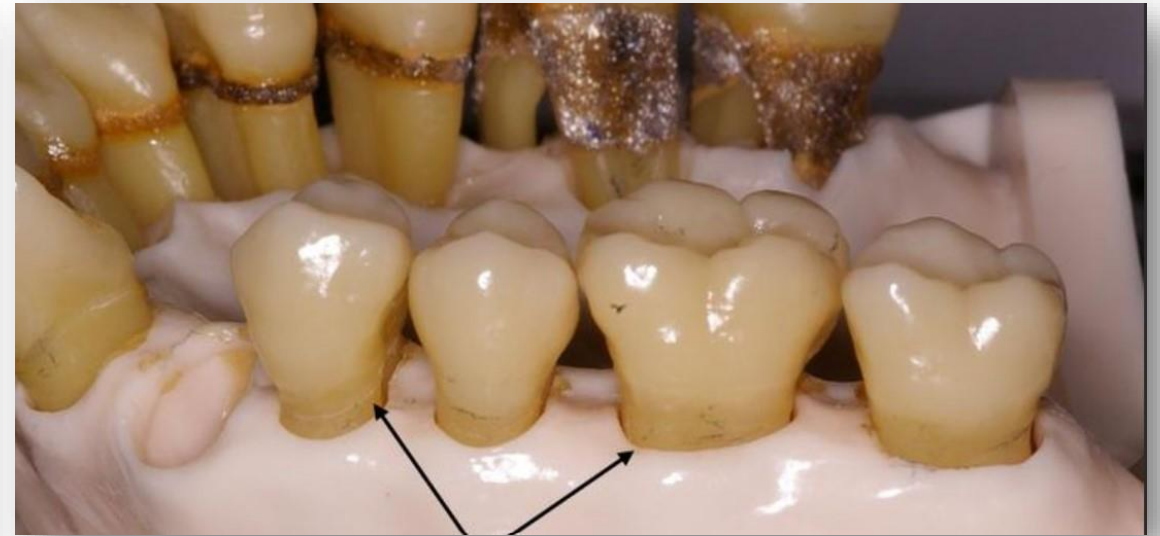
Identification of CEJ-explorer



• **Proximal CEJ is concave**



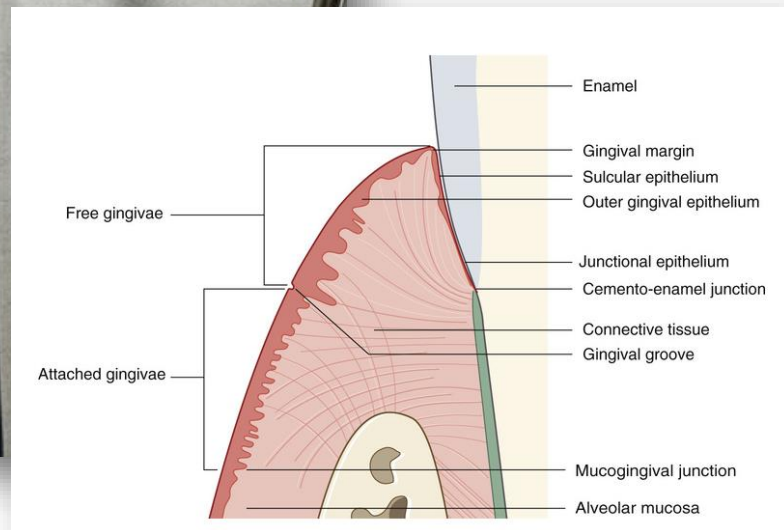
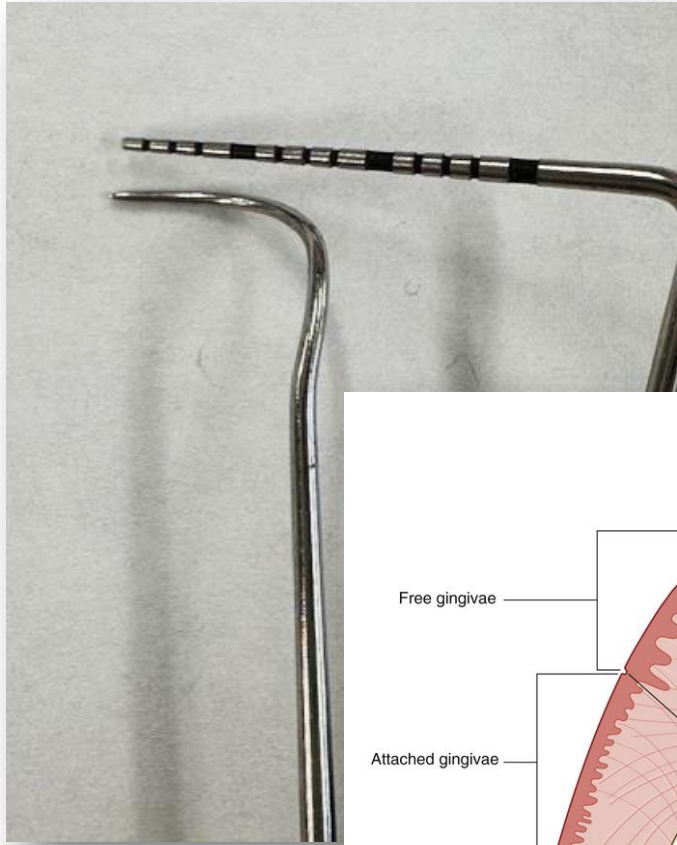
Facial and lingual CEJ is convex apically



ID CEJ Clinically



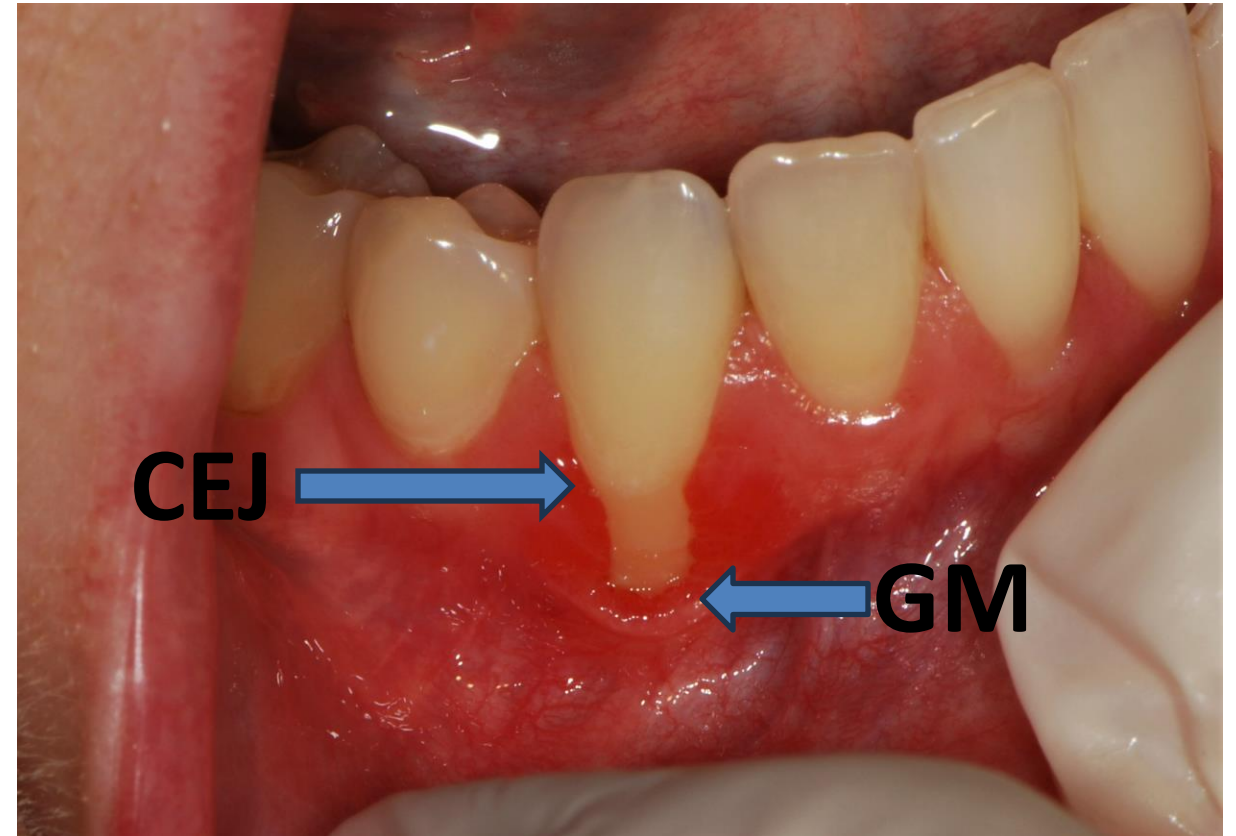
Measure GM to CEJ just like PDs (3 points on Buccal, 3 points on Lingual)



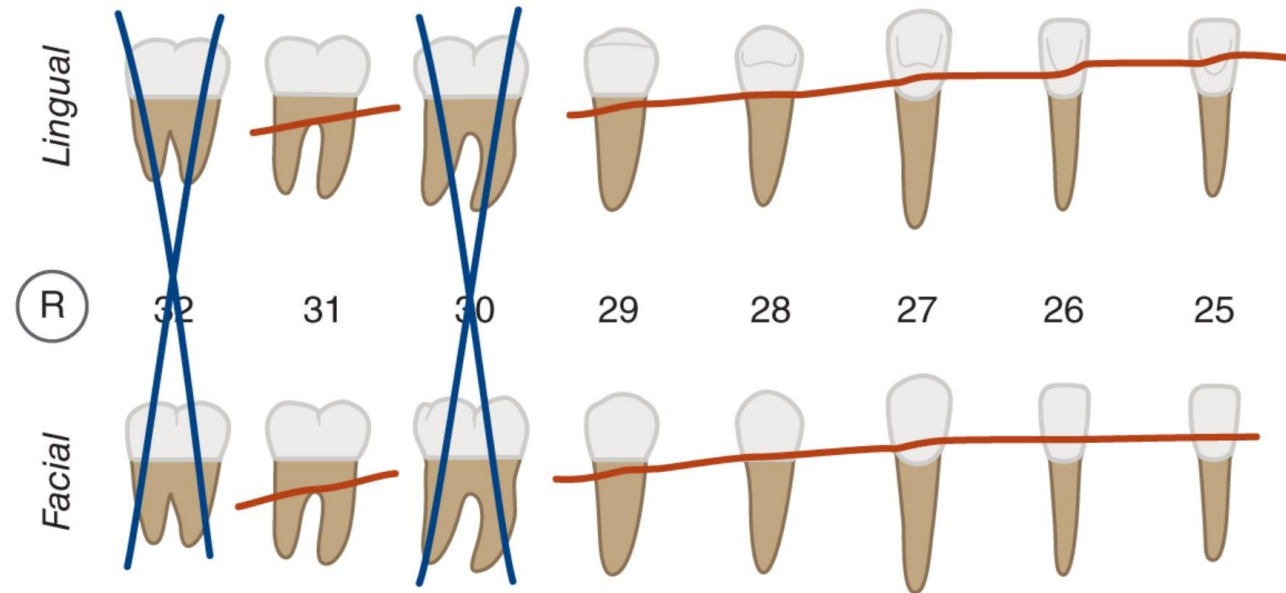
- In Health: GM is 1-3mm over the CEJ, mm is SUBTRACTED from PD to get CAL
- Gingivitis, Gingival Enlargement: Tissue over CEJ (negative mm)
- If GM is at CEJ...PD=CAL

Gingival Recession: GM to CEJ

- Apical migration of the gingival margin from CEJ
- Gingival Recession (GR) measured in millimeters (probe)
- GR=distance from GM to CEJ
- Added to PD to calculate CAL
- For example:
 $3\text{mm PD} + 2\text{mm GR} = 5\text{mm CAL}$



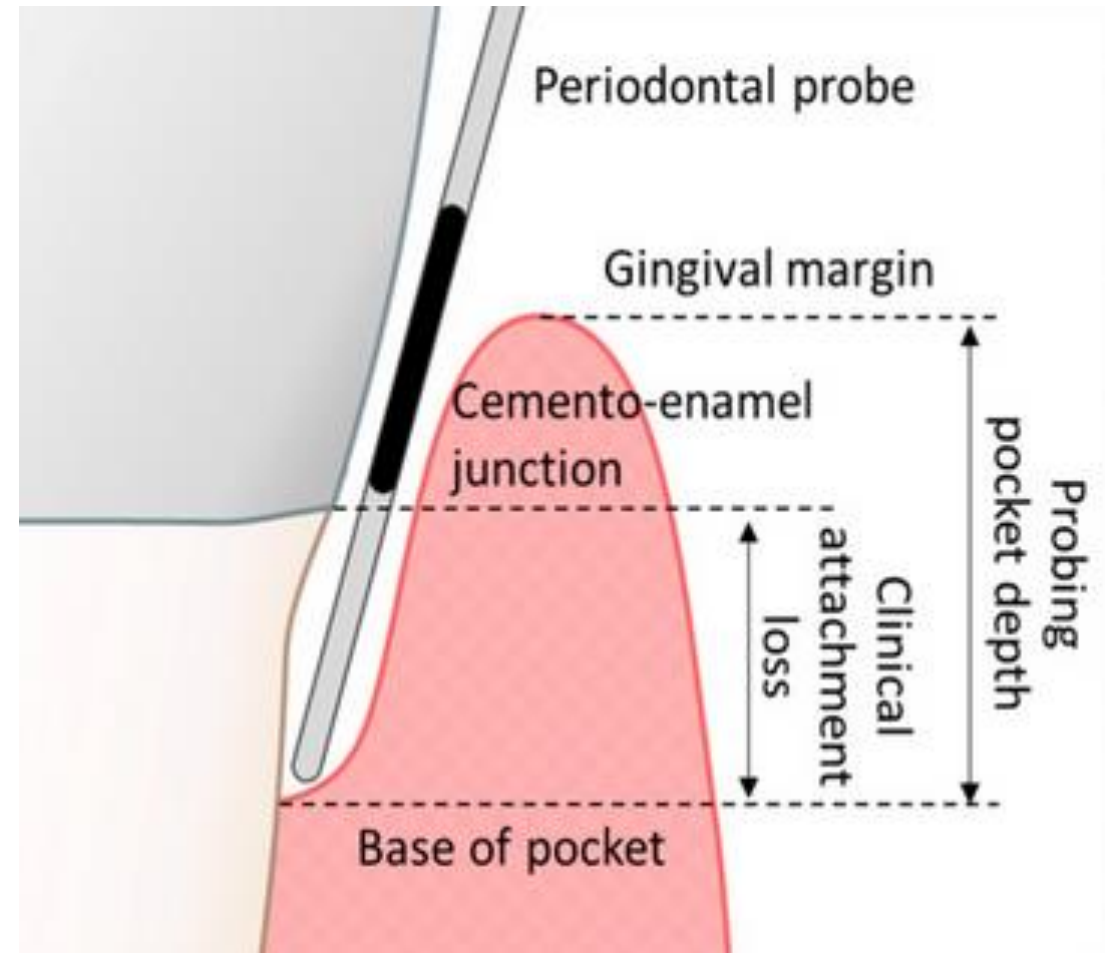
Probe depth		4 3 3		3 2 3	3 3 4	4 5 5	5 5 6	6 6 6
GM to CEJ		+5 +4 +3		+2 +2 +1	0 0 0	-1 -2 -2	-2 -2 -3	-3 -4 -4
Attachment loss		9 7 6		5 4 4	3 3 4	3 3 3	3 3 3	3 2 2



Probe depth		3 3 2		2 1 2	2 2 3	3 4 4	4 4 5	5 5 5
GM to CEJ		+4 +4 +3		+2 +2 +1	0 0 0	-1 -2 -2	-2 -2 -2	-2 -2 -2
Attachment loss		7 7 5		4 3 3	2 2 3	2 2 2	2 2 3	3 3 3
Mobility								

CAL vs. PD

- In health or gingivitis, it is common for the PD to be 2-4mm
- In health or gingivitis, it is normal for the GM to be coronal to (over) the CEJ 1-3mm
- If the PD is 4mm, and the tissue is 3mm over the CEJ, the CAL is 1mm



CAL: clinical attachment level/loss

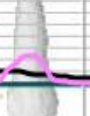
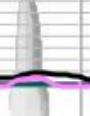
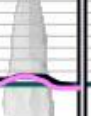
- More accurate indicator of periodontal support than only probing depth
- CAL is measured from a fixed point on a tooth, the CEJ
- CAL is the distance from the CEJ to base of the probing depth
- To calculate CAL, two measurements are needed:
 - In Recession: Probing depth (PD) + gingival recession (GR) = CAL (Add)
 - In Gingival Enlargement or Overgrowth: Probing depth(PD) – GE (measurement of tissue over the CEJ)=CAL (Subtract)
 - In Health or Gingivitis: GM is over the CEJ
 - The GM-CEJ is a negative number and subtracted from PD to get CAL. CAL should be 0.



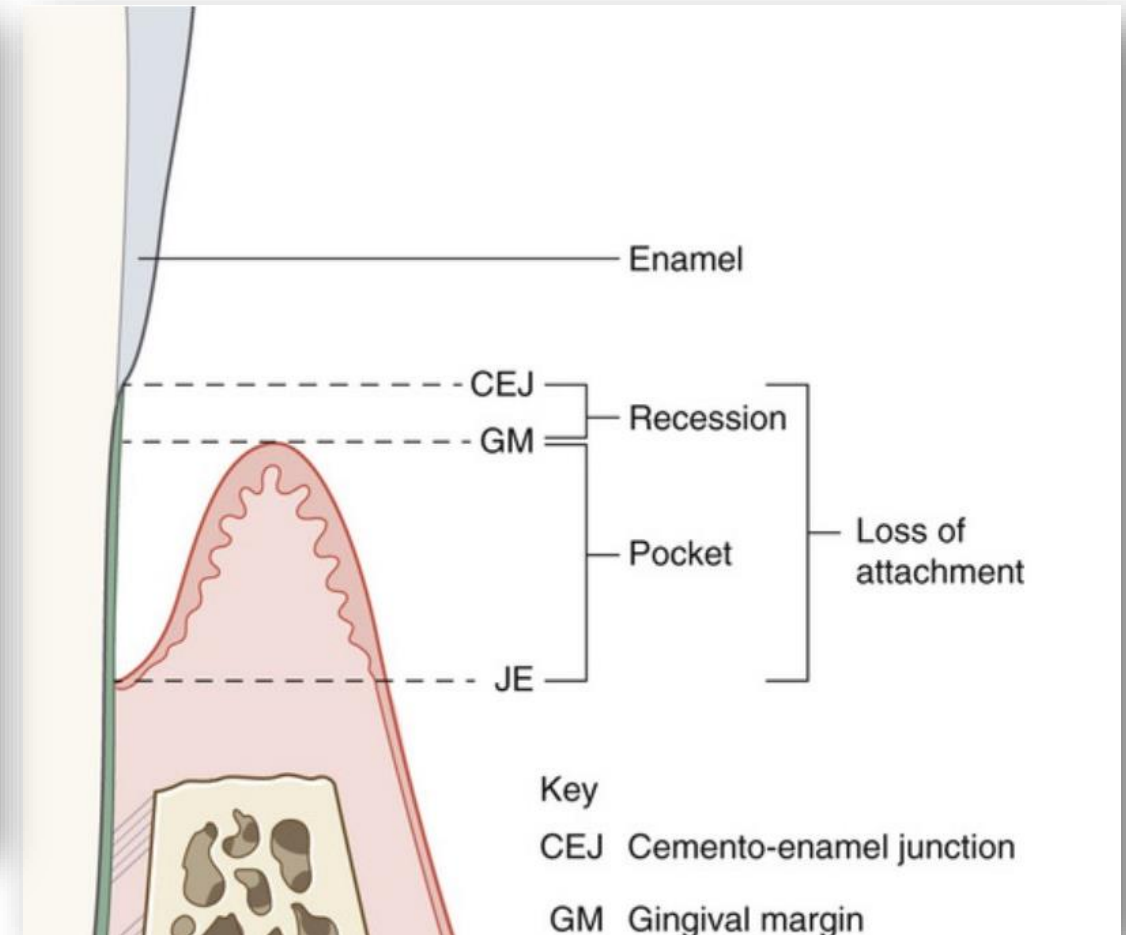
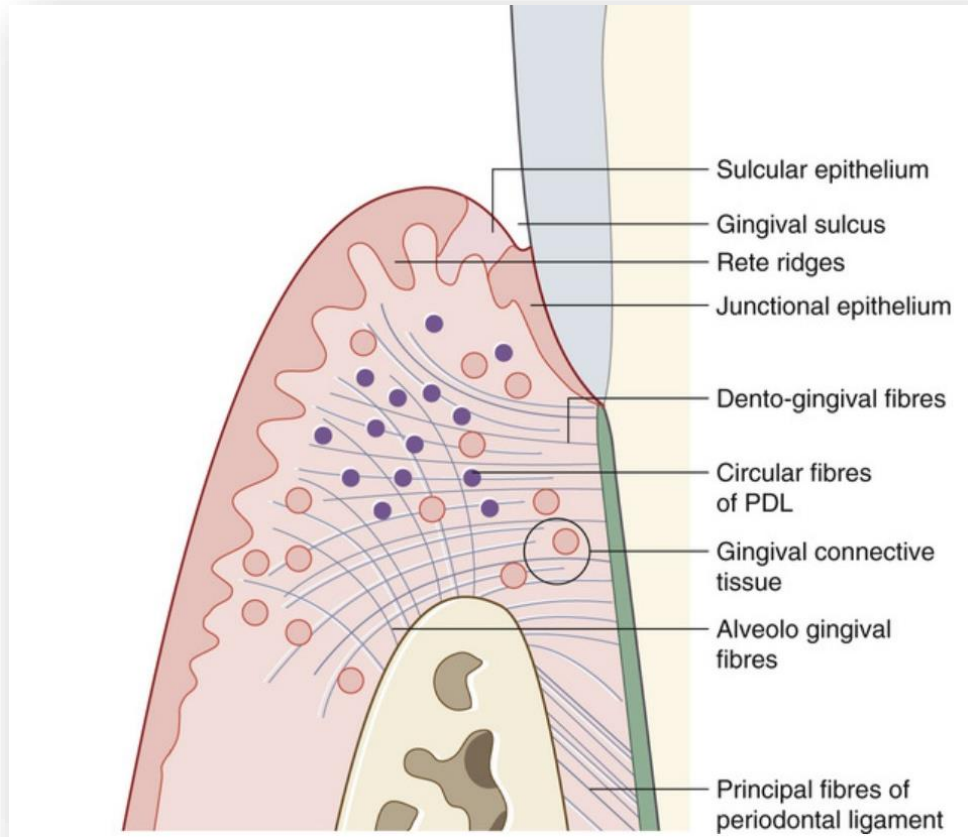
Gingival Enlargement (tissue overgrowth)

CAL in Health

- Health: Teeth #2&3 have 2-3mm PDs, No BOP, and no radiographic bone loss...we are assessing these teeth as having “0 mm of CAL”, so part of our calculation is to *estimate* that the GM to CEJ=-2to -3

P Depth		3 2 3	3 2 3	3 2 3	3 2 3	3 2 3	3 2 3	3 2 3	
GM CEJ		-3 -2 -3	-3 -2 -3			1 3 1	0 1 0	0 1 0	
Attach		0 0 0	0 0 0			4 5 4	3 3 3	3 3 3	
Calculus									
Plaque									
Suppuratn									
Bleed									
Furcation									
Facial	CEJ								
		1X	2	3	4	5	6	7	8
									

Health vs. Loss of Attachment



Tooth Mobility

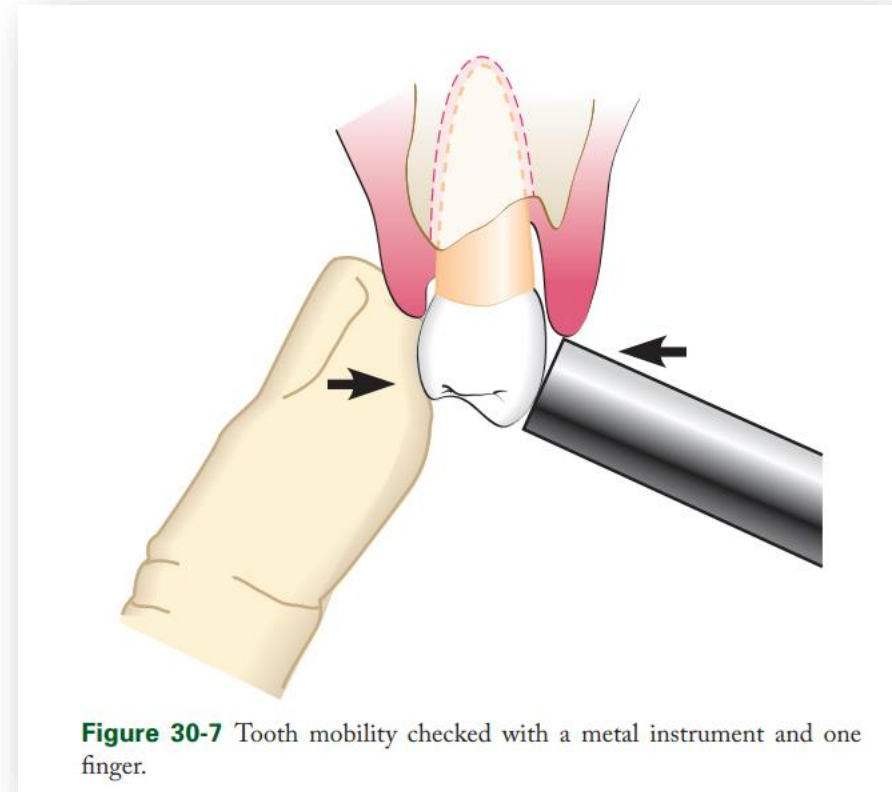


Figure 30-7 Tooth mobility checked with a metal instrument and one finger.

Tooth Mobility

Class I	Class II	Class III
Tooth can be moved less than 1mm in horizontal plane	Tooth can be moved 1-2mm in horizontal plane but cannot be moved vertically	Tooth can be moved in excess of 2mm in the horizontal plane, or in a vertical direction

Miller 1950

Occlusal Trauma

- Primary occlusal trauma: injury resulting in tissue changes from traumatic occlusal forces applied to teeth with **normal periodontal support**. This manifests itself clinically with *adaptive mobility* and is not progressive
- Secondary occlusal trauma: has been defined as injury resulting in tissue changes from normal or traumatic occlusal forces applied to a tooth or teeth with **reduced support**.

- Teeth with *progressive mobility* may also exhibit migration and pain on function
- Current periodontal therapies are directed primarily to address etiology; in this context, traumatic occlusal forces (check occlusion)
- Fremitus:
- Teeth with progressive mobility may require splinting for patient comfort.



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What is a furcation?



- The anatomic area of a multi-rooted tooth where the roots diverge.

Furcation involvement on dry skull

class I



Incipient

class II



cul-de-sac

class III



thru-thru

Furcation Classification

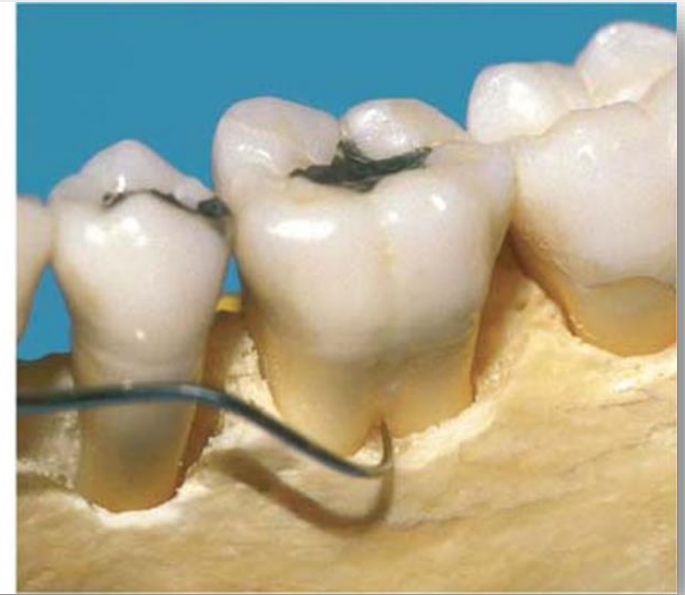
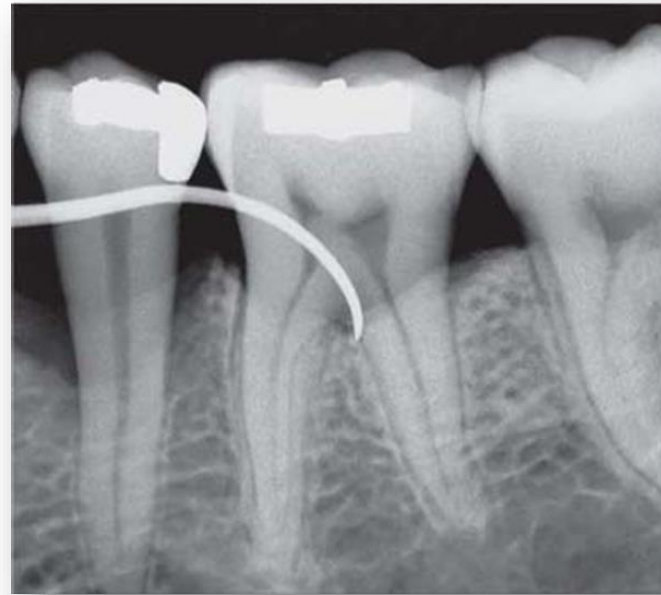
Glickman Classification (1953)

- **Grade 1:** primarily soft tissue and minimal bone loss in furcation (no radiographic signs)
- **Grade 2:** bone is destroyed in 1+ aspects of furca but not a “through and through” furcation; a periodontal probe will not pass through completely (radiographic signs can vary)
- **Grade 3:** bone is resorbed so that a probe could pass completely through; however, the furcation is not visible because of gingival coverage (bone loss is obvious on radiograph)
- **Grade 4:** visible furcation; both bone and soft tissue are missing (radiographically same as grade 3)

Hamp Furcation

- system devised by Hamp et al. (1975) uses the width of the tooth as the measure of furcation involvement
- Degree 1 – loss of horizontal periodontal support not exceeding one-third of the width of the tooth
- Degree 2 – loss of horizontal periodontal support exceeding one-third of the width of the tooth but not encompassing the total width of the furcation
- Degree 3 – a “through-and-through” furcation defect

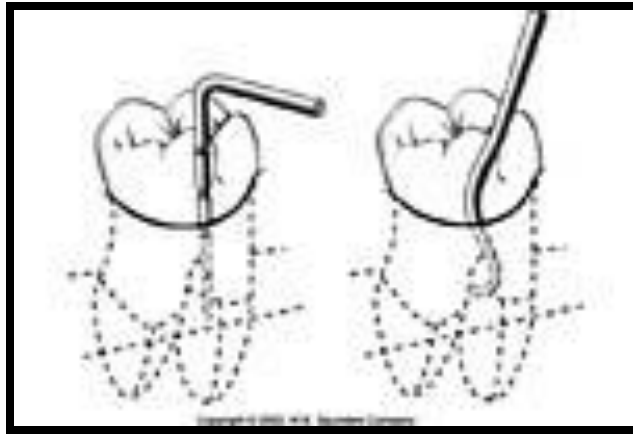
Nabors probe-furcation involvement



Sidebar: furcations, root morphology

Probing furcations- Nabers probe





This shows the position of the periodontal probe and the parallelism to the long axis of the tooth.



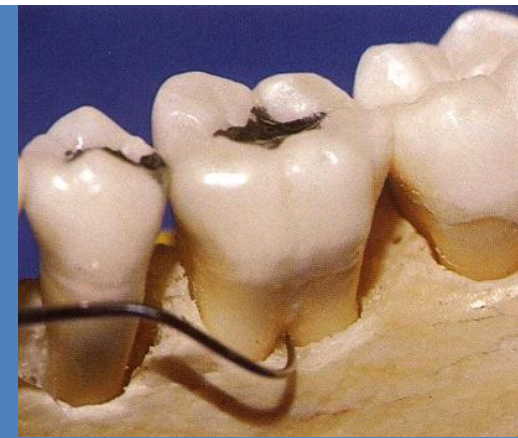
This shows a Nabers Probe detecting the furcation under the soft tissue.



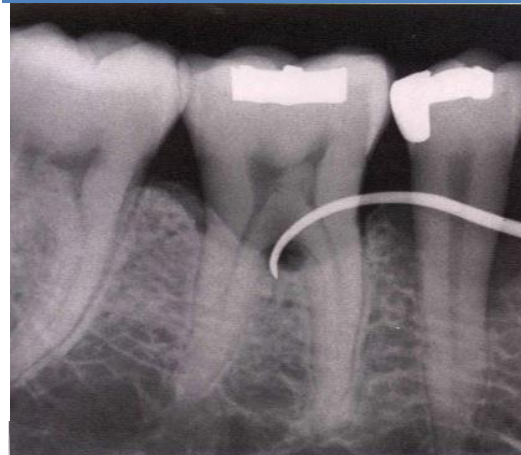
This shows the stages of furcation involvement

Furcation Classifications

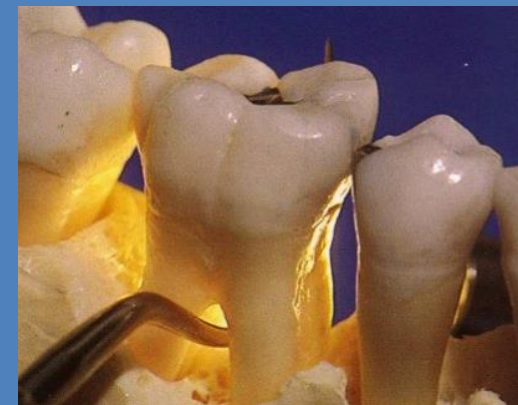
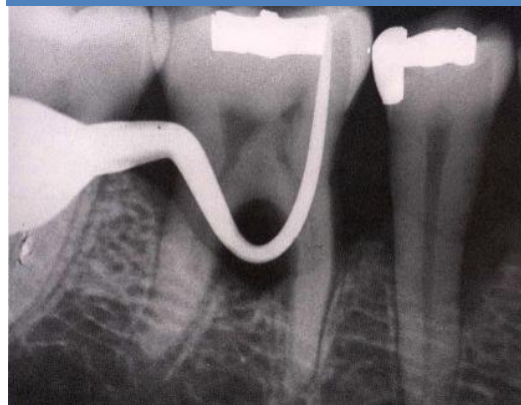
Class I



Class II

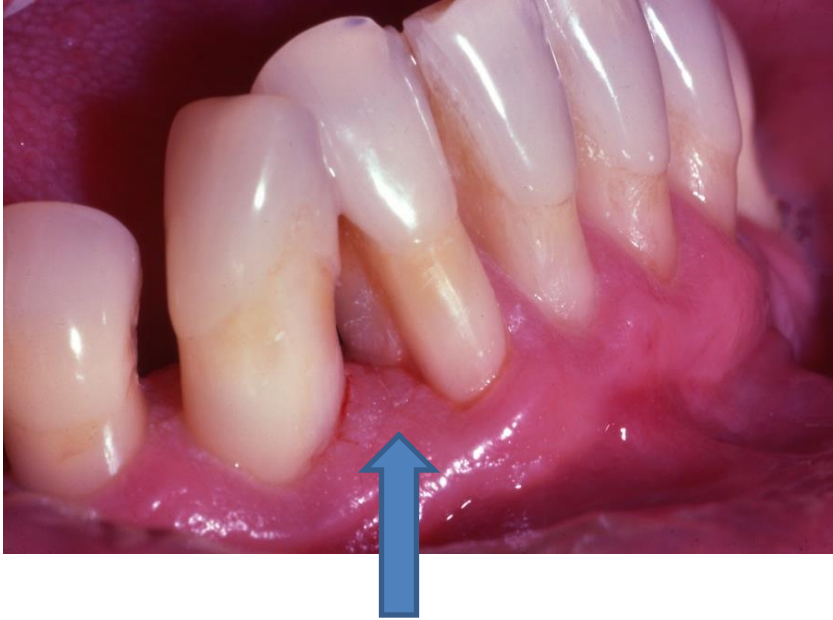


Class III

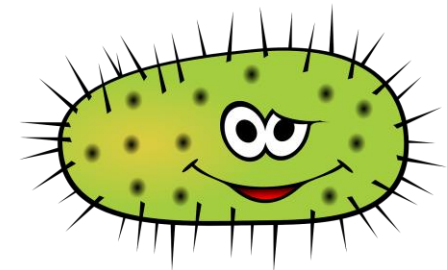


From Rateitshack, Wolf, and Hassel

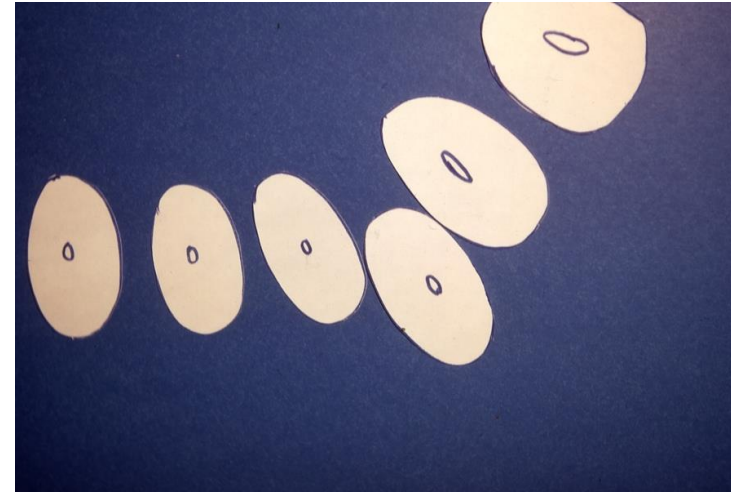
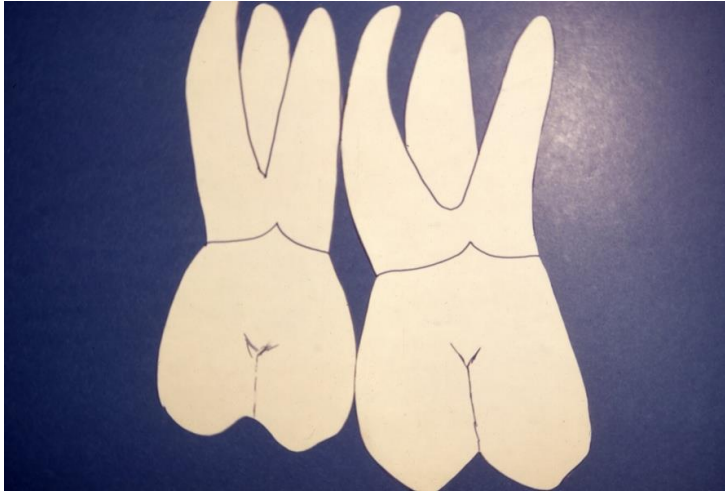
Root Morphology



Root concavities and furcations, “nooks and crannies”... hard to maintain, but bacteria love them



Root proximity or overcrowded teeth



Difficult access for plaque control by patient, debridement by professional, and surgical treatment such as bone grafting

Mandibular molar furcation

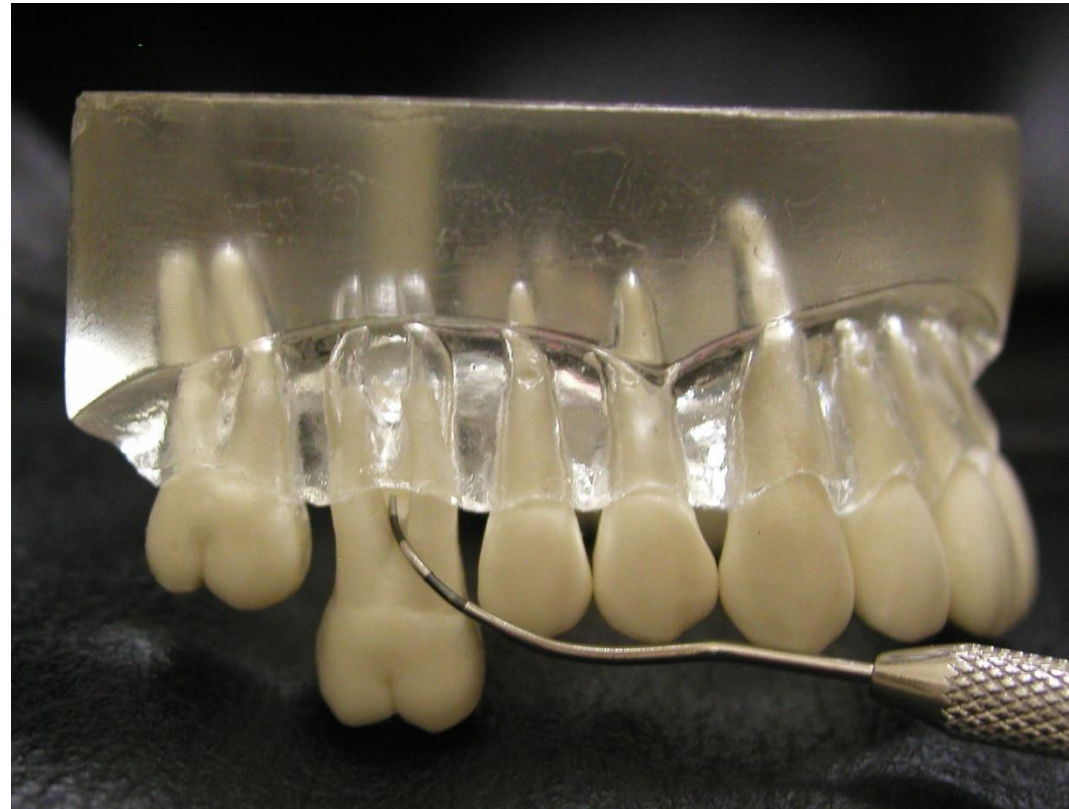


Detect mand molar lingual furcation

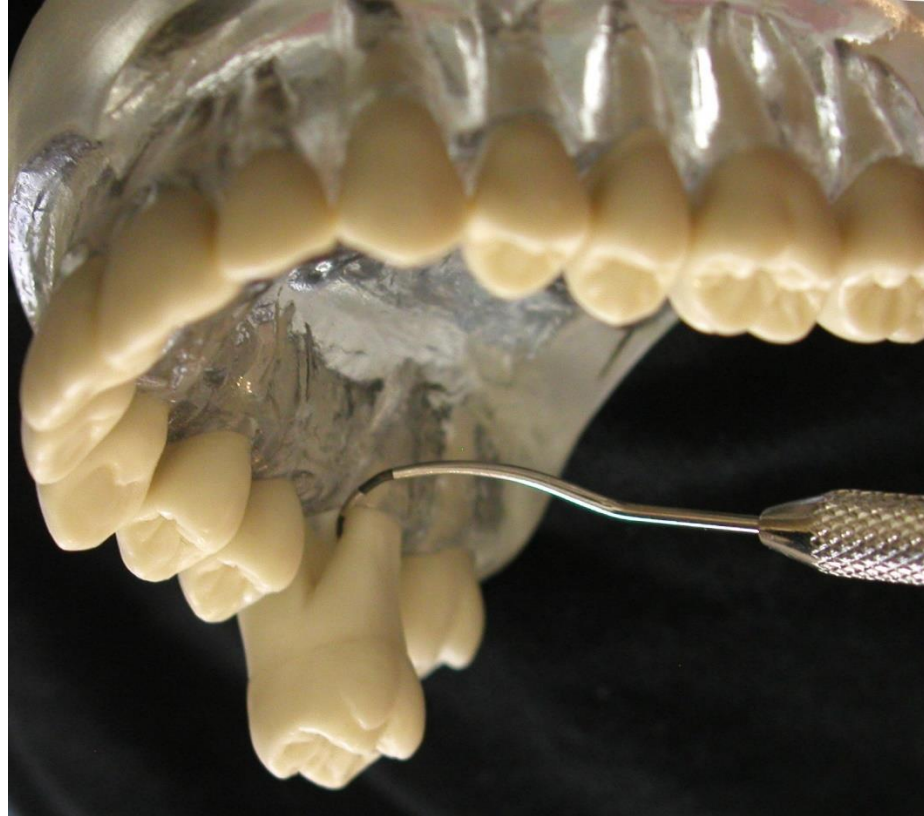


Lingual furcation has a narrower entrance and longer trunk root, making it harder to clean.

Detect maxillary molar buccal furcation

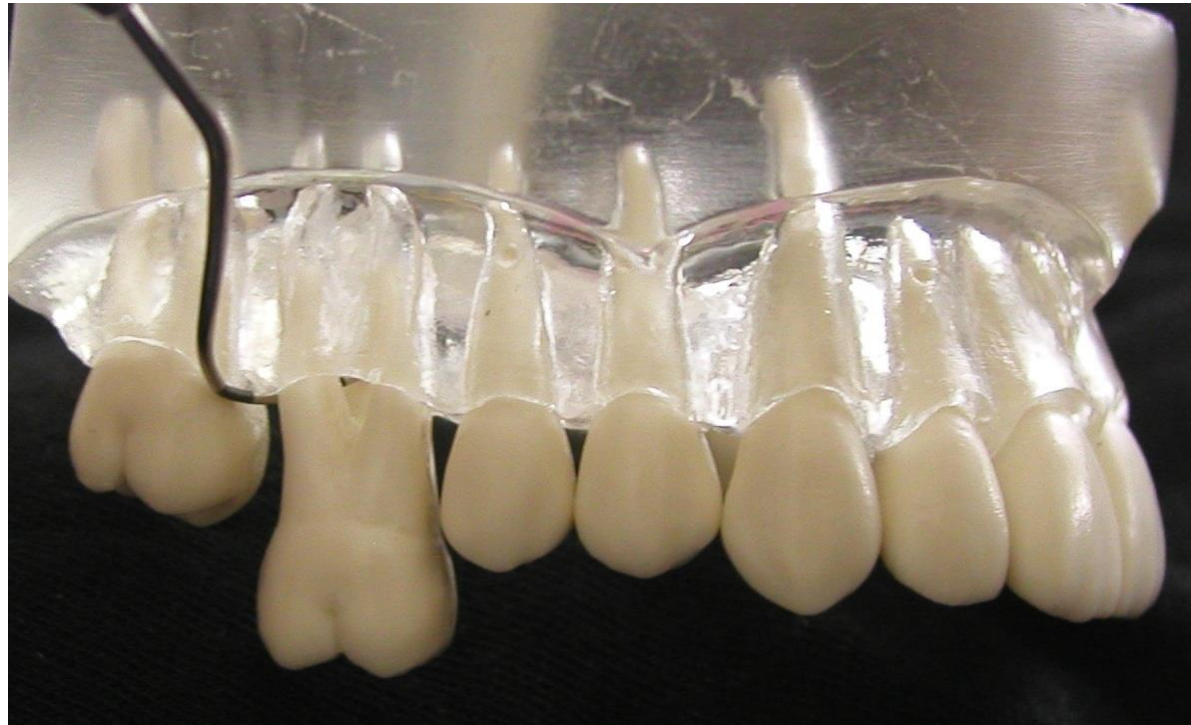


Detect maxillary molar mesial furcation

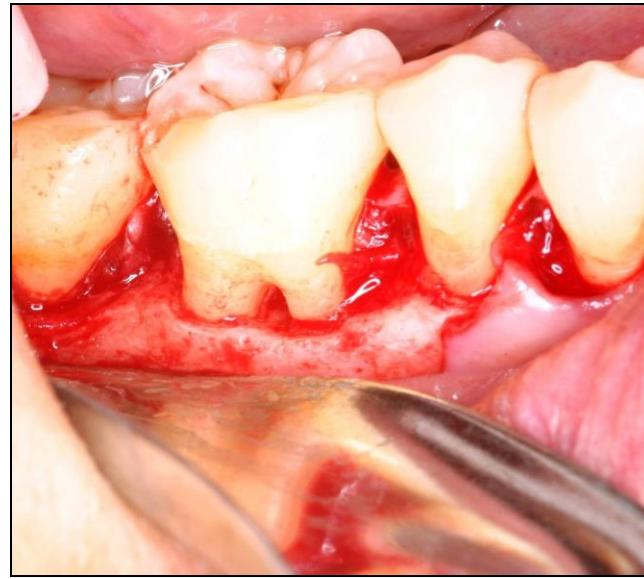


- Access from **palatal**!!!

Detect maxillary molar distal furcation



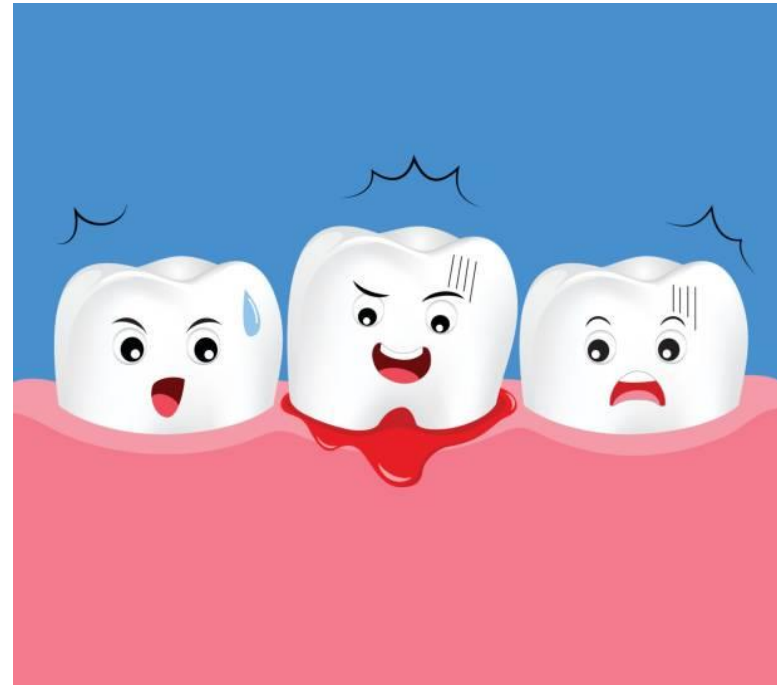
Radiographic and Surgical Exposure of a Furcation Defect



Direct visualization of a furcation is through surgery

Assessment of periodontal condition:

1. Symptoms of patient
2. Medical and Dental History
3. Visual Assessment
4. Periodontal Examination
5. Radiographic Assessment
6. Identification of etiology



5. Radiographic Assessment

Imaging techniques:

- Radiographs
- CBCT (cone beam computerized tomography)
- Photography

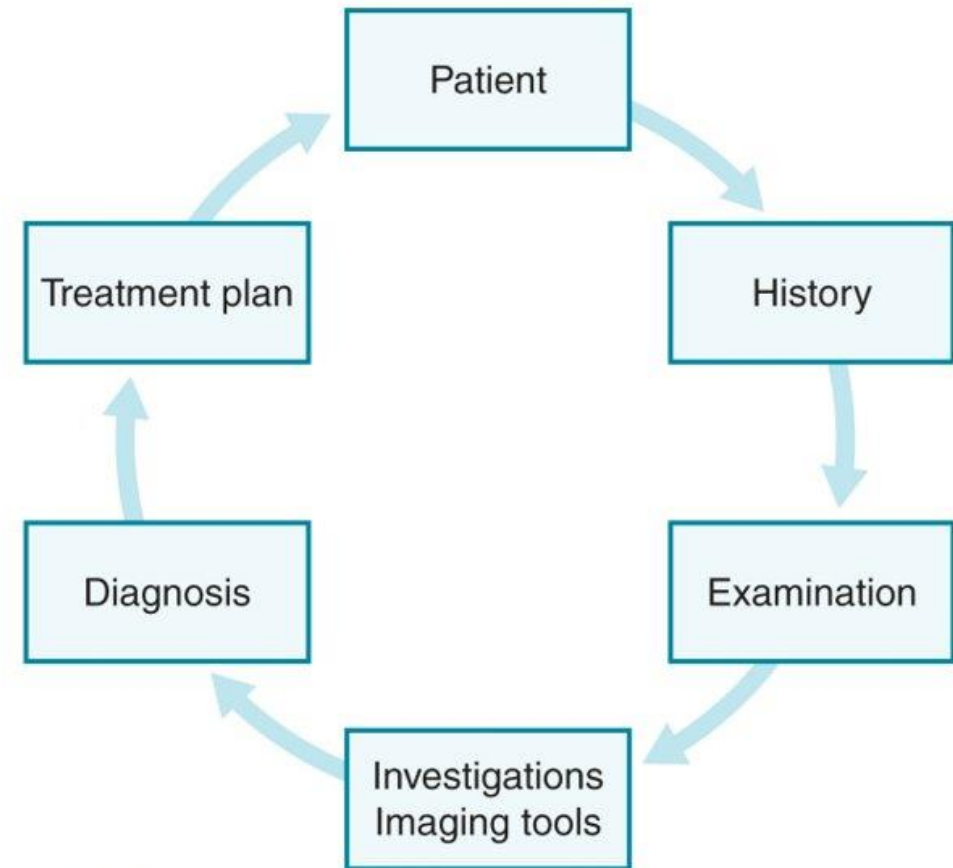


FIGURE 4.2.1 The place for imaging in patient management.

Learning Objectives

1. Understand and describe the components of a Periodontal Assessment and Examination
2. Understand how anatomy, biofilm, pathogenesis and host response influence the signs we see clinically
3. Understand probing depth vs. clinical attachment level/loss and the difference
4. Understand mobility, furcation, and plaque indices used